```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload the image
uploaded = files.upload()

# Get the uploaded filename
filename = next(iter(uploaded))

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Check if image loaded successfully
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Sobel X - detects vertical edges
    sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

    # Sobel Y - detects horizontal edges
    sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Convert to absolute 8-bit images
    sobelx = cv2.convertScaleAbs(sobelx)
    sobely = cv2.convertScaleAbs(sobely)

    # Combine Sobel X and Sobel Y
    edges = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

    # Display original image
    print("Original Image:")
    cv2_imshow(img)

    # Display Sobel X
    print("Sobel X:")
    cv2_imshow(sobelx)

    # Display Sobel Y
    print("Sobel Y:")
    cv2_imshow(sobely)

    # Display combined edge detection
    print("Combined Sobel Edge Detection:")
    cv2_imshow(edges)

    # Save the result
    cv2.imwrite("Edge_detection.jpg", edges)
```

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    print("Edge_detection.jpg saved successfully.")
```

Choose Files sampled_image.jpg

sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

```
Saving sampled_image.jpg to sampled_image (4).jpg
Original Image:
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Sobel X:
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Sobel Y:
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Combined Sobel Edge Detection:
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```
Edge_detection.jpg saved successfully.
```